

Day 4 — Why the Trick Works (Dot Drawing)

Module: Multiply-Near-Ten (Primary) · **Time:** 30 min

Goal

Children draw a dot pattern showing why $7 \times 8 = 56$ and see the connection between fingers and the answer.

Materials

- A4 paper per child (or larger).
- Coloured crayons / pencils (2 colours per child works fine).
- The hand poster from Day 3.

The flow

Settle + chant (3 min)

Quick Twins chant + 5 finger-trick chorus answers.

Story (7 min) — “Let’s see WHY”

- “Yesterday I told you the trick. Today I’ll show you why it works.”
 - Draw a 7×8 rectangle on the board: 7 columns, 8 rows.
 - Count the total dots together: 7 columns $\times$ 8 rows = 56 dots.
 - Now divide the rectangle into regions:
 - A **5×5 block** in one corner (the “fully solid” block).
 - A **5×3 strip** below (because $8 = 5 + 3$).
 - A **2×5 strip** to the side (because $7 = 5 + 2$).
 - A **2×3 corner** (fingers UP $\times$ fingers UP).
- Total dots: $25 + 15 + 10 + 6 = 56$. ✓
- “See the 2×3 corner? That’s the fingers UP — and $2 \times 3 = 6$, the units of our answer.”
 - “See the 5×5 block plus the two strips? Together they make 50 — the tens of our answer.”

Drawing (15 min)

- Each child draws a 7×8 dot rectangle on their A4 paper.
- Colour the 2×3 corner one colour (the “fingers up” region).
- Colour the rest a different colour (the “tens” region).
- Count and label: “fingers-up corner = ” **and** ”rest = .”

Show + chant (5 min)

- 2 children share their drawings.
- Final chant.

Homework

- Pick another problem (6×7 OR 9×8) and draw the dot rectangle. Colour the corners. Bring it tomorrow.

Differentiation

- **Tier up:** Draw the same picture for 9×9 . The corner is $1 \times 1 = 1$ dot. The rest is 80 dots. Total = 81. ✓
- **Tier down:** Pre-printed grid; child just colours the regions.

Teacher's quick notes

- The drawing IS the proof. Children who see the picture once will trust the finger trick forever.
- Some children will try to colour outside the lines, or count dots wrong. That's fine — the colouring is the point, the counting is the practice.
- Don't introduce algebra. The dot drawing is the entire "why" for this band.